DEEPTANSHU DEVATHA

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EDUCATION

Heritage High School, Frisco, TX — Class of 2027

GPA: 4.00 UW / 5.361 W · Rank **11/486** · ACT **36** · AP Scholar with Distinction

AP exams: 9 taken, a 5 on all 9 — Computer Science A, Statistics, Physics 1, Seminar, English Language & Composition, U.S. History, World History, Psychology, Human Geography

Coursework: AP Calculus BC, AP Physics 2, AP Physics 1, AP Computer Science A, AP English Language, AP English Literature, AP Government, AP Economics, Computer Science III, Mobile App Programming, Spanish 2

Honors: IEEE-EMBS IECBES 2026 paper accepted for presentation · X-Step startup: Conrad Innovator 2025–2026 — Health & Nutrition; Texas A&M McFerrin Center for Entrepreneurship Texas Ideas Challenge — Top 40 Finalist; Diamond Challenge 2026 — Semifinalist · ShopAI (as ShopSmart): TechLit Bridging the Gap Hackathon — 1st Place (Aug. 2025); PeddieHacks — Best Innovation Award, 1st/250 (Aug. 2025); Hack4Humanity — 3rd Place, 3/500 (Oct. 2025); AsterHacks — 5th Place, 5/250 (Oct. 2025)

RESEARCH

Predicting Cognitive Function Using Transformer-Derived Speech Representations and Longitudinal Coherence Features — *First Author* — May 2025–Present

Mentor: Joe Xiao (PhD student) · **Revision recommended by the editors, Frontiers in Aging Neuroscience — submitted August 04, 2026** · Preprint
medrxiv.org/content/10.64898/2026.08.04.26359730v1

- **Question:** can a patient's speech across clinic visits forecast their next Mini-Mental State Exam (MMSE) score? Framed as supervised regression on the DementiaBank Pitt Corpus (**173** longitudinal samples, **126** patients) using only earlier visits.
- **Representation:** to capture how discourse organization changes with decline, extracted 12 per-visit linguistic features (POS fragment ratio, named-entity rate, 50-token MATTR, filler rate, perseveration) and Sentence-BERT coherence features (adjacent-sentence cosine similarity, global coherence, semantic slope, topic drift, embedding variance), plus visit-to-visit Δ features.
- **Model:** reduced 384-d SBERT embeddings to 20 PCA components, added 9 clinical/demographic covariates (53 dimensions), and tuned LightGBM with Optuna/TPE over 200 trials (RobustScaler, correlation pruning, permutation-importance selection).
- **Evaluation:** found patient-level leakage (one patient on both sides of a split) and rebuilt evaluation with patient-grouped 5-fold GroupKFold, augmented copies kept in their patient's fold, original samples only scored: **R^2 0.840 ± 0.015**, **RMSE 2.75**, **MAE 2.02** MMSE points. Ablation: clinical history carries most of the signal (clinical-only **R^2 0.72**, SBERT-only **0.46**, linguistic-only **0.12**); speech adds a smaller gain (**0.76** combined); non-nested tuning is disclosed as an optimism limit. Code github.com/ddevatha1/dementia-mmse-prediction

ACCEPTED FOR PRESENTATION — IEEE-EMBS IECBES 2026 KUALA LUMPUR · DEC. 9–11, 2026

A Four-Site Plantar-Pressure Insole: Transducer Error Budget, Feature Equivariance, and Sparse-Versus-Dense Agreement in Overground Walking — *2nd of 5 Authors, ML Lead* — Jul. 2026–Present

UC Santa Cruz AIEA Lab (Prof. Leilani Gilpin) aiea-lab.github.io · deeptanshudevatha.com/research/x-step

- **Question:** can **four strategically placed sensing sites** capture regional pressure trends comparable to a **32-sensor insole**? I engineered the machine-learning component: a 59-feature representation of the team's four-site plantar-pressure readings (per-site peak/mean/CV, pressure–time integral, cadence, stance ratio, anterior load share) and a 540-parameter logistic-regression framework that classifies biomechanical risk.
- Designed the correlation study (**149 recordings**, **15 adults**, public 32-cell insole): four sites keep waveform shape (**median r = 0.72–0.99**) but not magnitude (**peak-to-peak r = 0.43–0.74**); center-of-pressure tracking keeps anteroposterior motion (**r = 0.81**) but loses mediolateral (**r = 0.01**).
- Quantified subject-identity leakage on a synthetic 24-subject cohort (**macro-F1 0.885** subject-grouped vs **0.931** ungrouped) and documented the generalization limit.

PROJECTS & LEADERSHIP

ShopAI — *Creator & Sole Developer* · renamed from ShopSmart; extends its core idea — 2025–Present

- Prices differ by store and no retailer offers an API built for comparison, so ShopAI searches nine chains at once and plans the trip. Built end to end: React Native/Expo client and an Express/TypeScript backend on AWS EC2 behind Nginx, with adapters that turn unlike sources (OAuth2 REST, cookie-session GraphQL, Playwright, server-rendered HTML) into one product schema. Live on the App Store. apps.apple.com/app/shopai/id6797083809
- Concurrent fan-out search under a shared response budget with per-store failure states, query-correction/relevance/ranking, and a per-ZIP auth warm-up state machine. A deterministic planner enumerates all **127** non-empty subsets of the 7 live-data stores (exhaustive search is cheap, so no heuristic) for cheapest, fastest, fewest-stop and weighted plans, with **zero** LLM calls. deeptanshudevatha.com/projects/shopai

KidCode — *Founder & Lead Instructor* · Frisco, TX — 2024–Present

- Founded a 3-year Python workshop serving **124 students** across 10 Frisco neighborhoods; recruited peers to help teach. Wrote the curriculum, **120 one-hour classes** (5/week, 8 weeks each summer) and **20 projects**: control flow, file I/O, JSON, exceptions, OOP/inheritance and assert-based tests, then spaCy NLP (tokenization, lemmatization, POS tagging, sentence segmentation, NER, Matcher patterns, a rule-based chatbot).
  deeptanshudevatha.com/projects/kidcode
- To test learning, wrote 13 pre/post assessment pairs (same skills, different code): across **104** students the mean score rose **41.7% → 58.8%** (**+17.1 pp**; **99** improved, 4 unchanged, 1 declined; no comparison group). Students completed **1,214** working programs, **572** using spaCy.
  deeptanshudevatha.com/projects/kidcode/assessment

X-Step — *Co-Founder, ML & Mobile Lead* — Aug. 2025–Present

- X-Step is a student startup building a smart insole for people at risk of diabetic foot ulcers, which form where pressure builds up unnoticed. The team's ESP32 insole streams four pressure sensors over BLE; I wrote the software: the React Native/Expo app (BLE client, live pressure heatmap, daily risk score, alerts, trends, guided routine) and the earlier Flask/SQLite + LLM health-assistant backend. deeptanshudevatha.com/projects/x-step
- Trained two CNN prototypes on public data: EfficientNet-B2 on 9 foot-pressure heatmap postures (**98.3%**, random 80/20 split) and ResNet-50 on four wound-severity grades. Intended flow: an ulcer photo's severity score plus care habits go to an LLM that drafts routines. Bench-stage prototype; no wearer data, no clinical validation.

Ayusha Nilayam — *Programming & Math Instructor* · Hyderabad, India — Nov. 2025–Jul. 2026

- Taught **38** weekly Zoom sessions of Python and math to **11** children at the Ayusha Nilayam orphanage in the Alwal/Kompally area of Hyderabad, adapting to one shared laptop, unreliable internet and a Telugu–English language gap; then two weeks in person. deeptanshudevatha.com/projects/ayusha-nilayam

UT Dallas NLP with Python Workshop — *Participant* — Summer 2025

Dr. Chris Davis · Built a rule-based text simplifier: dependency parsing, lexical substitution, clause splitting, passive-to-active rewriting.

TECHNICAL SKILLS

ML/NLP: LightGBM, scikit-learn, Optuna, Sentence-BERT, spaCy, PyTorch, PCA, GroupKFold, permutation importance · **Languages:** Python, Java, TypeScript/JavaScript, SQL
Technologies: React Native/Expo, Node.js/Express, Flask, REST, GraphQL, OAuth2, Playwright, AWS EC2, Nginx